REAL ESTATE CRM + WEBSITE ADMIN PANEL

Functional Requirements & Implementation Specification

Real Estate Consulting / Brokerage Business

Project Type: Real Estate CRM + Property Management + Website CMS

Primary Objective: Centralized CRM and website administration platform

Core Principle: CRM as the single source of truth for property inventory

1. PROJECT OBJECTIVE

- Lead and enquiry management
- Client management
- Owner management
- Property inventory management
- Residential property management
- Commercial property management
- Rental and leasing management
- Pre-leased properties
- Investment properties
- Plot and land management
- Project management
- Follow-up management
- Site visit management
- Sales pipeline management
- Rental/leasing pipeline management
- Staff and user management
- Dashboard and reporting
- Website content management
- Website property publishing
- Website enquiry capture
- Communication history
- WhatsApp integration where supported
- API and webhook integration
- Data import/export
- Role-based access control
- Activity/audit tracking
- Mobile-responsive CRM access

The core principle of the system is: CRM = Master Database / Single Source of Truth.

2. SYSTEM ARCHITECTURE

Recommended architecture: Admin/CRM Panel -> CRM Backend/API -> Central Database -> Website API -> Public Website.

Additional integration flows: Website Enquiry -> CRM Lead API; CRM Property -> Website Publishing API; CRM Status Change -> Website Update/Unpublish; CRM -> WhatsApp/Email/SMS/Notifications; CRM -> Third-party Property Portals.

3. SINGLE SOURCE OF TRUTH

The CRM database is the authoritative database for properties, projects, owners, clients, leads, staff, site visits, follow-ups, sales transactions, rental transactions and website-managed property information.

Example: a new property is created in CRM with Publish on Website = YES. It becomes available on the public website. Price, rent, description and availability changes are propagated to the website. Sold, Leased, Rented or Not Available status removes the property from active website listings while retaining its historical CRM record.

4. ADMIN PANEL

- Dashboard
- Leads
- Enquiries
- Clients
- Owners
- Properties
- Projects
- Residential Properties
- Commercial Properties
- Plots / Land
- Rental / Leasing
- Pre-Leased / Investment
- Site Visits
- Follow-ups
- Tasks
- Sales Pipeline
- Rental / Leasing Pipeline
- Investment Pipeline
- Bookings / Transactions
- Documents
- Communications
- Campaigns / Lead Sources
- Staff / Users
- Reports
- Website CMS
- Media Library
- Notifications
- Integrations
- Import / Export
- Audit Logs
- Master Settings
- System Settings

5. DASHBOARD

5.1 KPI Cards

- Total Leads
- New Leads
- Active Leads
- Converted Leads
- Lost Leads
- Today's Leads
- Today's Follow-ups
- Pending Follow-ups
- Overdue Follow-ups
- Today's Site Visits
- Upcoming Site Visits

- Total Properties
- Available Properties
- Sold Properties
- Rented Properties
- Leased Properties
- Pre-Leased Properties
- Investment Properties
- Website Published Properties
- Unpublished Properties
- Properties Added This Month
- Properties Sold This Month

5.2 Sales KPIs

- Total opportunities
- Active opportunities
- Negotiations
- Bookings
- Closed deals
- Conversion percentage
- Expected revenue
- Actual revenue
- Brokerage generated
- Brokerage pending
- Brokerage received

5.3 Team Performance

- Leads assigned per employee
- Leads contacted
- Follow-ups completed
- Follow-ups overdue
- Site visits completed
- Site visits upcoming
- Deals closed
- Revenue generated
- Conversion rate by employee

5.4 Lead Source Analytics

- Lead count
- Qualified lead count
- Site visits
- Conversions
- Revenue
- Cost/ROI where campaign data is available

6. LEAD & ENQUIRY MANAGEMENT

6.1 Lead Information

- Lead ID

- Lead Name
- First Name
- Last Name
- Mobile
- WhatsApp Number
- Alternate Mobile
- Email
- Lead Source
- Campaign
- Requirement Type
- Property Type
- Transaction Type
- Budget
- Preferred Location
- Preferred Area
- Minimum Area
- Maximum Area
- Bedrooms
- Furnishing Preference
- Parking Requirement
- Possession Requirement
- Purchase/Rental Purpose
- Investment Preference
- Assigned Employee
- Priority
- Lead Status
- Lead Stage
- Expected Closing Date
- Follow-up Date
- Notes
- Communication History
- Related Property
- Related Project
- Created Date
- Updated Date

6.2 Lead Types

- Buyer
- Tenant
- Investor
- Commercial Buyer
- Commercial Tenant
- Corporate Requirement
- Plot Buyer
- Property Seller
- Property Owner
- Landlord
- Developer

6.3 Lead Sources

- Website
- Google Ads
- Facebook Ads
- Instagram
- WhatsApp
- Referral
- Direct
- Phone
- Walk-in
- Property Portal
- Existing Client
- Broker
- Campaign

6.4 Lead Status

- New
- Contacted
- Qualified
- Requirement Captured
- Property Suggested
- Site Visit Planned
- Site Visit Completed
- Negotiation
- Booking
- Converted
- Not Interested
- Lost
- Future Follow-up

6.5 Lead Assignment

- Manual assignment
- Round-robin
- By location
- By property type
- By employee
- By team
- By availability
- Custom assignment rules

7. WEBSITE ENQUIRY -> CRM

Every website enquiry must be automatically captured in the CRM.

Flow: Website property page -> Enquire Now -> form submission -> Website -> CRM Lead API -> create/update lead -> assign sales employee -> notify employee/manager.

Duplicate prevention: search by mobile, email and WhatsApp number. If a match exists, update the existing lead and append the new enquiry activity; otherwise create a new lead.

8. CLIENT / CONTACT MANAGEMENT

- Client ID
- Name
- Mobile
- WhatsApp
- Email
- Address
- City
- Occupation
- Company
- Requirement
- Preferred Property Type
- Budget
- Preferred Location
- Transaction Type
- Documents
- Notes
- Related Leads
- Related Properties
- Site Visits
- Deals
- Communication History

Client 360° view should show basic information, requirements, suggested properties, interested properties, site visits, follow-ups, calls, WhatsApp messages, emails, deals, documents and activity timeline.

9. OWNER MANAGEMENT

- Owner ID
- Owner Name
- Mobile
- WhatsApp
- Email
- Address
- Company
- PAN/GST where applicable
- Preferred communication method
- Ownership type
- Notes
- Documents

Owner 360° view should show total properties, available/sold/rented/leased properties, enquiries, interested leads, site visits, transactions, brokerage, documents and communication history. One owner can own multiple properties.

10. PROPERTY INVENTORY MANAGEMENT

Core property categories:

- Residential: Apartment, Flat, Independent House, Villa, Row House, Bungalow, Builder Floor, Studio, Penthouse

- Commercial: Office, Retail Shop, Showroom, Commercial Building, Warehouse, Godown, Industrial Property, Factory, Co-working Space, Commercial Land
- Land: Residential Plot, Commercial Plot, Agricultural Land, Industrial Land, NA Plot, Farm Land
- Special Investment: Pre-Leased Property, Investment Property, Income Generating Property

11. PROPERTY BASIC DETAILS

- Property ID
- Property Title
- Property Type
- Property Category
- Transaction Type
- Residential/Commercial
- Listing Type
- Property Status
- Availability
- Featured Property
- Publish on Website
- Owner
- Assigned Employee
- Project
- Location
- Address
- Landmark
- City
- State
- Pincode
- Latitude
- Longitude

12. PROPERTY AREA DETAILS

- Configurable units: Sq. Ft., Sq. M., Sq. Yd., Acre, Hectare, Guntha, Bigha and other admin-defined units
- Carpet Area
- Built-up Area
- Super Built-up Area
- Plot Area
- Usable Area
- Frontage
- Depth
- Dimensions

13. APARTMENT / FLAT SPECIFIC DETAILS

- Project
- Tower/Block
- Floor
- Total Floors
- Flat Number

- Bedrooms
- Bathrooms
- Balcony
- Facing
- Furnishing
- Parking
- Lift
- Power Backup
- Security
- Age of Property
- Possession Status
- Construction Status
- Maintenance Charges
- Society Name
- Amenities

14. VILLA / HOUSE SPECIFIC DETAILS

- Plot Area
- Built-up Area
- Floors
- Bedrooms
- Bathrooms
- Kitchen
- Balcony
- Garden
- Terrace
- Parking
- Servant Room
- Furnishing
- Facing
- Age
- Construction Status
- Possession
- Amenities

15. PLOT / LAND MANAGEMENT

- Plot ID
- Plot Title
- Plot Type
- Residential/Commercial
- Plot Area
- Area Unit
- Frontage
- Depth
- Dimensions
- Road Width
- Facing

- Corner Plot
- Gated Layout
- NA Status
- Authority
- Layout Name
- Survey Number
- Plot Number
- Zone
- Location
- Landmark
- Google Maps Location
- Ownership
- Price
- Price Per Sq. Ft.
- Price Per Sq. Yd.
- Negotiable
- Documentation Status
- Water Availability
- Electricity Availability
- Road Connectivity
- Nearby Facilities
- Description
- Photos
- Videos
- Documents
- Website Publish Status
- Availability

16. COMMERCIAL PROPERTY MANAGEMENT

- Office, Shop, Showroom, Warehouse, Factory, Godown, Retail, Industrial, Corporate Space, Commercial Land
- Carpet Area
- Built-up Area
- Floor
- Ceiling Height
- Parking
- Furnishing
- Power Load
- Washrooms
- Lift
- Fire Safety
- Security
- Maintenance
- Rent
- Sale Price
- Deposit
- Lock-in
- Lease Period

- Escalation
- Possession
- Current Tenant
- Current Lease Status
- Brokerage Terms

17. RENTAL / LEASING MANAGEMENT

17.1 Rental

- Monthly Rent
- Security Deposit
- Maintenance
- Notice Period
- Available From
- Furnishing
- Lease Period
- Tenant
- Owner
- Brokerage
- Rental Status

17.2 Leasing

- Lease Start
- Lease End
- Lease Duration
- Lock-in Period
- Escalation
- Renewal Date
- Rent Escalation Date
- Current Rent
- Security Deposit
- Tenant
- Landlord
- Brokerage
- Agreement Document

18. PRE-LEASED / INVESTMENT PROPERTY

- Property
- Tenant
- Lease Start
- Lease End
- Current Rent
- Expected Annual Rent
- Security Deposit
- Lease Escalation
- Lock-in
- Yield

- Capital Value
- ROI
- Remaining Lease
- Tenant Type
- Tenant Company
- Investment Status
- Expected Return
- Investment Notes

Calculated metrics should include annual rental income, rental yield, expected ROI and remaining lease period.

19. PROJECT MANAGEMENT

- Project ID
- Project Name
- Developer
- Project Type
- Location
- Address
- City
- Landmark
- RERA Number where applicable
- Construction Status
- Launch Date
- Possession Date
- Total Units
- Available Units
- Sold Units
- Amenities
- Project Description
- Project Images
- Project Videos
- Brochure
- Master Plan
- Floor Plans
- Website Publish Status

Projects may contain towers, blocks, buildings, floors and units. Project-level and unit-level inventory should both be supported.

20. UNIT INVENTORY

- Unit Number
- Unit Type
- Floor
- Facing
- Carpet Area
- Built-up Area
- Balcony Area
- Price
- Parking

- Availability
- Booking Status
- Customer
- Booking Date

Unit statuses: Available, Hold, Booked, Sold, Cancelled, Blocked.

21. PROPERTY SEARCH & FILTERING

Internal CRM search: Property Type, Transaction Type, Location, Area, Budget, Bedrooms, Furnishing, Floor, Parking, Status, Owner, Project, Property ID, Assigned Staff, Investment Type, Rental Yield.

Website search: Buy, Rent, Lease, Commercial, Residential, Plots, Location, Price, Area, Property Type, Bedrooms, Amenities, Availability.

22. LEAD-TO-PROPERTY MATCHING

Match customer requirements against available inventory using location, property type, budget, area, bedrooms, transaction type, availability, furnishing and parking. Allow sales users to select and share matching properties by WhatsApp, email or shareable public link.

23. SALES PIPELINE

1. New Lead
2. Contacted
3. Qualified
4. Requirement Captured
5. Property Suggested
6. Property Shortlisted
7. Site Visit Scheduled
8. Site Visit Completed
9. Negotiation
10. Offer Submitted
11. Booking
12. Agreement
13. Sale Closed
14. Lost

Each stage should retain stage entry date, responsible employee, mandatory information, next follow-up and notes/activity history.

24. RENTAL / LEASING PIPELINE

15. New Enquiry
16. Qualified
17. Requirement Captured
18. Property Suggested
19. Shortlisted
20. Visit Scheduled
21. Visit Completed
22. Negotiation

23. Documentation
24. Agreement
25. Lease Started
26. Closed
27. Lost

25. INVESTMENT PIPELINE

28. New Investor
29. Requirement Captured
30. Investment Properties Shared
31. Property Shortlisted
32. Financial Evaluation
33. Site Visit
34. Negotiation
35. Due Diligence
36. Booking
37. Agreement
38. Investment Closed
39. Lost

26. FOLLOW-UP MANAGEMENT

- Lead/Client
- Assigned Employee
- Date
- Time
- Follow-up Type
- Subject
- Notes
- Priority
- Status
- Outcome
- Next Follow-up

Types: Call, WhatsApp, Email, Meeting, Site Visit, Document Collection, Negotiation, Reminder, Other. Statuses: Pending, Completed, Rescheduled, Cancelled, Missed.

27. TASK MANAGEMENT

- Personal tasks
- Lead tasks
- Property tasks
- Site visit tasks
- Documentation tasks
- Management tasks
- Task Title
- Description
- User
- Related Lead

- Related Property
- Due Date
- Priority
- Status
- Reminder

28. SITE VISIT MANAGEMENT

- Visit ID
- Lead
- Client
- Property
- Assigned Employee
- Date
- Time
- Meeting Point
- Site Address
- Status
- Attendance
- Customer Feedback
- Employee Notes
- Outcome
- Next Follow-up

Statuses: Scheduled, Confirmed, Rescheduled, Completed, Cancelled, No Show. Optional future features: GPS check-in/out, distance tracking, site visit proof and customer digital confirmation.

29. BOOKING / TRANSACTION MANAGEMENT

- Transaction ID
- Client
- Property
- Owner
- Employee
- Transaction Type
- Sale/Rent/Lease
- Agreed Price
- Rent
- Deposit
- Brokerage
- Brokerage Percentage
- Brokerage Amount
- Payment Status
- Agreement Status
- Booking Date
- Closing Date
- Documents

30. BROKERAGE / COMMISSION MANAGEMENT

- Sale brokerage
- Rental brokerage
- Leasing brokerage
- Employee commission
- Broker commission
- Partner commission

System should calculate brokerage based on fixed amount, percentage, or configured commercial rules.

31. DOCUMENT MANAGEMENT

- Property documents
- Sale deed
- Agreement
- PAN
- Aadhaar where legally appropriate
- RERA documents
- Lease agreement
- Rent agreement
- NOC
- Brochure
- Floor plan
- Site plan
- Property photos
- Videos
- Documents linked to Client, Owner, Property, Project, Lead, Deal, Tenant and Transaction

32. COMMUNICATION MANAGEMENT

- Phone
- WhatsApp
- Email
- SMS
- Website enquiry
- Internal notes
- Meetings

Customer communication should be visible in the customer activity timeline wherever the integration permits.

33. WHATSAPP INTEGRATION

- Send WhatsApp messages
- Receive messages where supported
- Send property details
- Send property images
- Send brochure
- Send meeting/site-visit confirmation
- Send follow-up reminders
- Send enquiry acknowledgement

- Send property recommendations
- Track communication history

Integration depends on a supported WhatsApp Business API/provider setup.

34. WEBSITE <-> CRM INTEGRATION

34.1 Website -> CRM

- New enquiry
- Contact form
- Property enquiry
- Callback request
- Site visit request
- Property requirement form
- General enquiry
- Investment enquiry
- Rental enquiry

34.2 CRM -> Website

- New property
- Updated property
- Property price
- Rent
- Description
- Photos
- Videos
- Amenities
- Location
- Availability
- Featured status
- SEO data
- Project information

35. PROPERTY PUBLISHING SYSTEM

Each property must include Publish on Website = YES/NO.

Publication conditions can require Publish = YES, Status = Active/Available and required public fields completed.

A property must be removed from active website listings when Publish = NO, or when Sold, Leased, Rented, Blocked, Not Available or Archived, subject to configured business rules.

36. PROPERTY STATUS AUTOMATION

Example: Available -> Sold. The system changes the CRM status, removes the property from active website listings, retains the CRM history, updates related deal/pipeline records, writes an activity log and records the status change date. Apply equivalent logic to Available -> Leased/Rented and other configured statuses.

37. WEBSITE CMS

- Home Page: hero banner, heading, subheading, CTA, featured properties, featured projects, services, about section, testimonials, statistics, featured locations and promotional banners
- About Us: company description, mission, vision, team, images, certifications and achievements
- Services: sales, purchase, rental, leasing, commercial, investment consultancy, property management, land/plot consultancy
- Projects: title, description, location, developer, amenities, images, videos, brochure, floor plans, project status, units, price range, website visibility
- Locations: city, area, locality, landmark, description, SEO content and related properties
- Testimonials: customer name, designation, photo, review, rating and publish status
- Blog/Articles: title, slug, category, author, content, featured image, SEO metadata, publish date and status
- FAQ: question, answer, category, display order and status
- Contact Page: phone, email, WhatsApp, address, map, working hours and social links

Admin should be able to add, edit, delete, reorder and enable/disable applicable CMS content.

38. WEBSITE MEDIA LIBRARY

- Property photos
- Project photos
- Videos
- Brochures
- Floor plans
- Documents
- Website banners
- Logos
- Team photos
- Blog images
- Upload, replace, delete, preview, search, categorize and associate media with property/project/page

39. WEBSITE SEO MANAGEMENT

- Page title
- Meta title
- Meta description
- SEO keywords
- Slug
- Canonical URL
- OG title
- OG description
- OG image
- Sitemap inclusion
- Index/No-index
- Schema fields where applicable

40. WEBSITE PROPERTY DETAIL PAGE

- Property title
- Property ID

- Photos
- Video
- Price
- Rent
- Area
- Bedrooms
- Bathrooms
- Floor
- Furnishing
- Parking
- Location
- Map
- Amenities
- Description
- Owner/listing information where appropriate
- Availability
- Enquiry CTA
- Schedule Visit CTA
- WhatsApp CTA
- Call CTA
- Similar properties

41. WEBSITE SEARCH

- Buy
- Rent
- Lease
- Commercial
- Residential
- Plot
- Location
- Price
- Area
- Bedrooms
- Property Type
- Furnishing
- Amenities
- Availability

42. FEATURED PROPERTY

Support Featured = YES/NO. Featured properties can appear on the home page, featured sections, recommendations and configured landing pages. Selection is controlled from CRM/admin.

43. PROPERTY RECOMMENDATION

Provide rule-based recommendations using location, property type, budget, area, bedrooms, transaction type and availability. Future enhancement: AI property recommendations and lead scoring.

44. PROPERTY PORTAL INTEGRATION

- 99acres
- MagicBricks
- Housing.com
- Other supported portals

Provide a common export/feed/API layer so external property portals can receive approved inventory without duplicate manual data entry.

45. LEAD IMPORT

- Excel import
- CSV import
- Field mapping
- Duplicate detection
- Validation
- Error report
- Import history

46. PROPERTY IMPORT

- Bulk property import
- Field mapping
- Validation
- Duplicate detection
- Import history
- Import fields for property ID, type, location, area, price, rent, owner, status, website publish and description

47. DATA EXPORT

- Leads
- Clients
- Owners
- Properties
- Projects
- Site visits
- Follow-ups
- Deals
- Transactions
- Users
- Reports
- Excel/CSV and PDF where applicable
- Filtered exports

48. STAFF / USER MANAGEMENT

- Employee ID
- Name

- Email
- Mobile
- Department
- Role
- Designation
- Branch
- Profile photo
- Joining date
- Status
- Login status

49. ROLE-BASED ACCESS CONTROL

Role	Typical Access
Super Admin	Full system access
Admin	Operational access
Sales Manager	Team leads, properties, follow-ups, site visits, pipelines, reports
Sales Executive	Assigned leads, properties, follow-ups, site visits, communication
Property Manager	Property inventory, owners, tenants, lease information
Website Manager	Website CMS, properties, projects, media, blog, SEO
Management / Viewer	Dashboard and reports

Permissions should support View, Create, Edit, Delete, Export, Assign, Publish and Approve.

50. LEAD VISIBILITY

- Only assigned leads
- Team leads
- All company leads

Visibility must be configurable and enforced by role/permission.

51. APPROVAL WORKFLOW

Optional controlled publication workflow: Property Draft -> Verification -> Approved -> Published; Project Draft -> Review -> Approved -> Published; Website Content Draft -> Review -> Published.

52. NOTIFICATION SYSTEM

- New Lead
- Lead Assignment
- New Website Enquiry
- Follow-up Reminder
- Overdue Follow-up
- Site Visit Reminder
- Site Visit Cancellation
- Property Approval
- Property Published
- Property Sold

- Property Leased
- New Task
- Task Due
- Deal Closing
- Lease Expiry

Channels: in-app, email, WhatsApp where configured, and push notifications for a mobile application where implemented.

53. AUTOMATION ENGINE

Support configurable rule-based automation.

Example - New Lead: Website source -> create lead -> assign user -> create follow-up -> notify sales manager.

Example - Site Visit Tomorrow: notify employee and notify customer where communication is configured.

Example - Property Sold: set Publish on Website = NO -> remove from active website results -> update related pipeline -> record audit entry.

Example - Overdue Follow-up: mark overdue -> notify assigned employee -> escalate to manager when enabled.

54. WORKFLOW / BLUEPRINT ENGINE

Stage-based processes should support controlled transitions, mandatory fields, required notes, next follow-up and activity history. Example: New -> Qualified -> Site Visit -> Negotiation -> Booking -> Closed.

55. CAMPAIGN MANAGEMENT

- Campaign Name
- Campaign Type
- Start Date
- End Date
- Budget
- Source
- Target Property
- Location
- Leads Generated
- Qualified Leads
- Site Visits
- Conversions
- Revenue

Example campaigns: Google Ads, Facebook Ads, Instagram, Property Portal, WhatsApp, Email and Referral.

56. CALL MANAGEMENT

- Click to call
- Incoming call logging where supported
- Outgoing call logging
- Call duration
- Call outcome

- Call notes
- Next follow-up

57. REPORTING MODULE

57.1 Lead Reports

- Leads by date
- Leads by source
- Leads by employee
- Leads by location
- Leads by property type
- Converted leads
- Lost leads

57.2 Property Reports

- Available properties
- Sold properties
- Rented properties
- Leased properties
- Properties by location
- Properties by type
- Properties by owner

57.3 Site Visit Reports

- Visits by employee
- Visits by property
- Completed visits
- Cancelled visits
- No-shows

57.4 Sales Reports

- Closed deals
- Deal value
- Brokerage
- Employee performance
- Monthly sales

57.5 Rental Reports

- Active rentals
- Lease expiries
- Rent amount
- Deposit
- Tenant information

57.6 Website Reports

- Published properties
- Unpublished properties
- Website enquiries
- Enquiries by property

- Most-viewed properties where analytics are integrated

58. AUDIT LOG

Every important action should record user, action, entity, entity ID, old/new values when relevant, timestamp and source.

Actions: Create, Update, Delete, Publish, Unpublish, Assign, Status change, Login, Logout, Export and Import.

59. LOGIN & SECURITY

- Secure authentication
- Password hashing
- Session management
- Role permissions
- Account activation/deactivation
- Password reset
- Login history
- API authentication
- Rate limiting
- Input validation
- File validation
- Access control
- Optional 2FA
- Optional device/session management
- Optional IP restrictions

60. API ARCHITECTURE

Secure REST API groups should cover authentication, leads, properties, website/public properties, enquiries, projects, site visits, follow-ups, clients, owners, users, reports and website CMS.

Example endpoints: POST /api/auth/login; GET/POST/PUT/DELETE /api/leads; GET/POST/PUT/DELETE /api/properties; GET /api/public/properties; GET /api/public/properties/:slug; GET /api/public/projects; GET /api/public/locations; GET /api/public/properties/featured; POST /api/public/enquiries; GET/POST /api/projects; GET/POST /api/site-visits; GET/POST /api/followups.

61. WEBHOOK ARCHITECTURE

- property.created
- property.updated
- property.published
- property.unpublished
- property.sold
- property.leased
- lead.created
- lead.updated
- site_visit.created
- site_visit.updated

Webhooks must support secure authentication/signatures, delivery status, retry and logging.

62. WEBSITE API

- GET /api/public/properties
- GET /api/public/properties/:slug
- GET /api/public/projects
- GET /api/public/locations
- GET /api/public/properties/featured
- POST /api/public/enquiries

Only approved public information may be exposed. Owner private data, internal notes, brokerage, employee information and internal documents must never be returned through public APIs.

63. WEBSITE ADMIN CONTENT FLOW

Admin Panel -> Create/Edit Content -> Validate -> Save in CRM/central database -> API -> Website. No separate manual website database entry should be required for property listings.

64. PROPERTY CONTENT VALIDATION

Before publication, validate mandatory fields such as property title, property type, transaction type, location, price/rent, description, main image and availability. Display a clear error describing missing fields when publication is blocked.

65. PROPERTY DUPLICATE CONTROL

- Same owner + address
- Same property number
- Same unit number
- Same survey/plot number
- Same project + tower + unit

Admin may override duplicate warnings when multiple legitimate listings represent the same physical asset.

66. WEBSITE PROPERTY URL / SLUG

Create an SEO-friendly URL for each property, for example /property/3-bhk-apartment-dharampeth-nagpur. Slugs should be editable by authorized users while the immutable property ID remains the stable internal reference.

67. PROPERTY PHOTOS

- Multiple images
- Cover image
- Image ordering
- Image captions
- Replace image
- Delete image
- Preview
- Compression/optimization
- Optional watermark
- Optional automatic resizing

- Optional WebP conversion

68. PROPERTY VIDEO

- Video URL
- Uploaded video where supported
- YouTube/Vimeo links
- Property walkthrough
- Virtual tour link

69. GOOGLE MAPS / LOCATION

- Address
- Latitude
- Longitude
- Google Maps URL
- Landmark

Public map display must follow privacy rules and configured public-location policy.

70. MOBILE CRM

- Desktop
- Laptop
- Tablet
- Mobile
- View leads
- Call clients
- WhatsApp clients
- Update lead stage
- Add notes
- Schedule follow-ups
- View properties
- Share properties
- View site visits
- Update site visit status
- View tasks
- Manage assigned enquiries

71. GLOBAL SEARCH

Single search should search Leads, Clients, Owners, Properties, Projects, Site Visits and Transactions. Examples: a person name returns matching contacts/leads/owners; a Property ID returns the property directly.

72. FILTERS

- Search
- Date filters
- Status filters

- Employee filters
- Location filters
- Type filters
- Price filters
- Custom filters
- Saved filters

73. DATA OWNERSHIP

The client should retain ownership of business data subject to the final contract. The platform should provide database backup, data export, CSV/Excel export, API access and document download/migration support.

74. BACKUP & RECOVERY

- Automated database backup
- Backup retention
- Media backup
- Disaster recovery procedure
- Restore testing
- Backup monitoring
- Daily incremental backup where supported
- Periodic full backup

75. MASTER DATA MANAGEMENT

- Property types
- Transaction types
- Property statuses
- Lead sources
- Lead statuses
- Locations
- Amenities
- Furnishing options
- Roles
- Other configurable master values

Master values should be configurable rather than hard-coded wherever practical.

76. WEBSITE FORMS

- Contact Us
- Request Callback
- Property Enquiry
- Schedule Visit
- Buy Property
- Rent Property
- Sell Property
- List Property
- Investment Enquiry

Each form submission should create or update the appropriate CRM enquiry/lead record.

77. OWNER PROPERTY SUBMISSION

Future-ready workflow: owner submits listing -> Property Submission -> Pending -> Verified -> Approved -> Published. Unverified owner listings must not be publicly published.

78. CUSTOMER PROPERTY REQUEST

- Name
- Mobile
- Email
- Buy/Rent
- Property Type
- Budget
- Location
- Area
- Bedrooms
- Additional Requirement

The request automatically becomes a structured CRM lead/requirement profile.

79. PROPERTY SHARING

- WhatsApp message
- Email message
- Public property link
- PDF property sheet
- Brochure
- Select multiple properties and share

80. PROPERTY COMPARISON

Parameter	Property A	Property B	Property C
Type	Apartment	Villa	Apartment
Area	1200 sq.ft	1800 sq.ft	1450 sq.ft
Price	₹55L	₹80L	₹68L
Bedrooms	3	3	3
Parking	1	2	1

81. CLIENT REQUIREMENT PROFILE

- Requirement
- Property type
- Location
- Budget
- Bedrooms
- Area

- Parking
- Furnishing
- Other custom requirements

This structured requirement profile drives property matching and follow-up workflows.

82. OWNER REQUIREMENT / SELLER MANAGEMENT

- Expected Price
- Minimum Acceptable Price
- Property Details
- Availability
- Preferred Buyer Type
- Brokerage
- Urgency
- Documentation Status

83. NOT AVAILABLE / ARCHIVE SYSTEM

Avoid unnecessary physical deletion of historical properties. Use statuses such as Active, Sold, Rented, Leased, Archived and Temporarily Unavailable so history and reporting are preserved.

84. ACTIVITY TIMELINE

Example lifecycle: Property Created -> Assigned to Employee -> Website Published -> Customer Enquiry -> Lead Created -> Site Visit Scheduled -> Negotiation -> Sold. Store a complete chronological event history against relevant entities.

85. MANAGEMENT REPORTS

- Monthly lead growth
- Lead conversion
- Sales value
- Rental value
- Brokerage
- Employee performance
- Property inventory
- Lead-source performance
- Site visit conversion
- Pending follow-ups
- Lost leads
- Top-performing properties
- Most active locations
- Website enquiry performance

86. ADMIN SETTINGS

- Company information
- Logo

- Address
- Phone
- Email
- WhatsApp
- GST details
- Social media
- Notification settings
- Email settings
- WhatsApp settings
- SMS settings
- Website settings
- SEO settings
- API keys
- Webhooks
- Backup settings
- User roles
- Permissions
- Master data

87. API DOCUMENTATION & HANDOVER

- Authentication
- Endpoint
- Method
- Request body
- Response
- Validation
- Error codes
- Pagination
- Filtering
- Sorting
- Webhooks
- Authentication keys
- Integration instructions

The client must not be dependent on the original implementation team for every future integration.

88. ERROR HANDLING

All APIs should return standardized success/error structures with meaningful machine-readable codes and field-level validation errors.

89. AUDITABLE INTEGRATION

- Event ID
- Event type
- Property ID/entity ID
- Timestamp
- Sync status
- Request metadata

- Response metadata
- Error
- Retry count

90. WEBSITE SYNC FAILURE & RETRY

A CRM update must never silently fail to reach the website. On failure, log the error, retry automatically, retain synchronization status and notify administrators after repeated failures.

91. RECOMMENDED SYNCHRONIZATION DESIGN

Although the requirement is called two-way synchronization, the recommended business rule is: CRM -> Website for master property/content synchronization, and Website -> CRM for lead/enquiry/customer input. This prevents both systems from independently overwriting the same property record.

92. CMS + CRM COMBINATION

Recommended final system: CRM + Property Management + Website CMS + Automation Engine + API Layer + Notification/Communication Layer + Reporting Layer.

93. FUTURE EXTENSIONS

- Mobile App
- AI Lead Scoring
- AI Property Recommendations
- AI Chatbot
- WhatsApp AI Agent
- Property Portal Sync
- IVR
- Call Recording
- Payment Gateway
- Online Booking
- Commission Settlement
- Rental Management
- Tenant Management
- Digital Agreements
- E-signature
- Accounting Integration
- Google Calendar
- Email Marketing
- Advanced Marketing Automation
- Advanced BI Dashboard

94. RECOMMENDED ROLE OF CRM + AUTOMATION

Use a combined architecture where the CRM/property-management system owns business data and the automation layer handles workflow, webhooks, communication, scheduled jobs and external integrations.

Standard CRM capabilities should be used where they fit; custom modules/functions and APIs should handle real-estate-specific requirements.

95. FUNCTIONAL MODULE SUMMARY

Module	Required
Admin Dashboard	Yes
Lead Management	Yes
Enquiry Management	Yes
Client Management	Yes
Owner Management	Yes
Staff Management	Yes
Roles & Permissions	Yes
Property Management	Yes
Apartment/Flat	Yes
Villa/House	Yes
Commercial	Yes
Plot/Land	Yes
Rental	Yes
Leasing	Yes
Pre-Leased	Yes
Investment Properties	Yes
Project Management	Yes
Unit Inventory	Yes
Sales Pipeline	Yes
Rental/Leasing Pipeline	Yes
Investment Pipeline	Yes
Site Visits	Yes
Follow-ups	Yes
Tasks	Yes
Lead-Property Matching	Yes
Transactions/Bookings	Yes
Brokerage/Commission	Yes
Documents	Yes
Communications	Yes
WhatsApp Integration	Yes / API dependent
Website CMS	Yes
Website Integration	Yes
Property Publishing	Yes
Automatic Unpublishing	Yes
SEO Management	Yes
Media Library	Yes
Reports	Yes
Notifications	Yes
Import/Export	Yes
Audit Logs	Yes
API	Yes
Webhooks	Yes
Property Portal Integration	Integration-ready
Mobile Responsive CRM	Yes
Advanced Automation	Yes
AI	Future-ready
Backup/Recovery	Yes

97. CRITICAL END-TO-END ACCEPTANCE TEST

TEST 1 - PROPERTY

Admin logs in -> creates apartment -> adds price/location/images -> sets Publish on Website = YES -> property appears on public website.

TEST 2 - PROPERTY UPDATE

Admin changes price, e.g. ₹60L -> ₹65L -> website displays ₹65L.

TEST 3 - SOLD

Admin changes status to SOLD -> property disappears from active website listing -> property remains in CRM history.

TEST 4 - WEBSITE ENQUIRY

Customer opens property -> submits enquiry -> CRM creates/updates lead -> lead is assigned -> employee receives notification.

TEST 5 - SALES PROCESS

Lead created -> requirement captured -> matching properties -> property shared -> site visit -> completed -> negotiation -> booking -> sale closed -> property becomes Sold -> website automatically unpublishes property.

98. FINAL PRODUCT DEFINITION

A complete Real Estate CRM and Website Management Platform that enables a real-estate consulting company to manage leads, clients, owners, property inventory, projects, plots, apartments, commercial properties, rental/leasing properties, investment properties, sales pipelines, follow-ups, site visits, staff, communication, reporting and the public website from one centralized admin system.

Architecture rule: CRM/property management is the single source of truth for property inventory, while the public website consumes approved property/content data through APIs. Website enquiries flow back into CRM as leads/enquiries.

99. IMPLEMENTATION DELIVERABLES

99.1 Application

- Admin CRM
- Property Management
- Website CMS
- Website Integration
- Dashboard
- Reports

99.2 Technical

- Production backend
- Production frontend
- Database
- APIs
- API documentation
- Webhook documentation
- Integration documentation

99.3 Data

- Import templates
- Export capability
- Backup procedure

99.4 Security

- Roles
- Permissions
- Authentication
- Audit logs

99.5 Handover

- Admin credentials
- Technical documentation
- API documentation
- Deployment documentation
- Database backup procedure
- Integration configuration documentation
- Basic administrator training

100. SCOPE COMPLETION CHECKLIST

[ ] Login & authentication
[ ] Dashboard
[ ] Lead management
[ ] Enquiry management
[ ] Client management
[ ] Owner management
[ ] Staff management
[ ] Roles & permissions
[ ] Property management
[ ] Apartment management
[ ] Villa/house management
[ ] Commercial management
[ ] Plot/land management
[ ] Rental management
[ ] Leasing management
[ ] Pre-leased management
[ ] Investment management
[ ] Project management
[ ] Unit inventory
[ ] Property search
[ ] Lead-property matching
[ ] Sales pipeline
[ ] Rental pipeline
[ ] Investment pipeline
[ ] Follow-ups
[ ] Tasks
[ ] Site visits
[ ] Transactions
[ ] Brokerage
[ ] Documents
[ ] Communications
[ ] Notifications
[ ] Campaigns
[ ] Reports
[ ] Import
[ ] Export
[ ] Audit logs
[ ] Website CMS
[ ] Home page management
[ ] About management
[ ] Services management
[ ] Project management

[ ] Website property management
[ ] Location management
[ ] Blog management
[ ] FAQ management
[ ] Testimonial management
[ ] Media library
[ ] SEO management
[ ] Website property API
[ ] Website enquiry API
[ ] CRM -> Website property publishing
[ ] CRM -> Website property updating
[ ] CRM -> Website property unpublishing
[ ] Website -> CRM enquiry capture
[ ] Duplicate lead prevention
[ ] API authentication
[ ] Webhooks
[ ] Sync logging
[ ] Retry mechanism
[ ] Mobile responsive interface
[ ] Security testing
[ ] Permission testing
[ ] Production deployment
[ ] Backup
[ ] API documentation
[ ] Technical handover
[ ] Admin handover